Maxwell–Teleparallel Algebra & Chern–Simons Theory

Reference Notes

TQFT–CFT Correspondence

WZW Model

G compact simply-connected simple, Σ closed Riemann surface, $k \in \mathbb{Z}_{>0}$:

$$S_{\mathrm{WZW}}[g] = \frac{k}{8\pi}\int_\Sigma \mathrm{tr}(g^{-1}\partial g)^2\, d^2x + \frac{ik}{12\pi}\int_B \mathrm{tr}(g^{-1}dg)^3.$$

Level quantisation: e^{iS} is single-valued since $H = \frac{1}{6}\mathrm{tr}(\theta^3) \in H^3(G,\mathbb{Z})$ (Maurer–Cartan form $\theta = g^{-1}dg$). Equations of motion: $\bar\partial J = 0$, $\partial \bar J = 0$ with holomorphic current $J = -k\,\partial g \cdot g^{-1}$.

Kac–Moody algebra. Mode expansion $J^a(z) = \sum_n J^a_n z^{-n-1}$ gives

$$[J^a_m, J^b_n] = if^{ab}{}_c J^c_{m+n} + km\,\delta^{ab}\delta_{m+n,0} \quad (\hat{\mathfrak{g}}_k).$$

Sugawara construction: $T(z) = \frac{1}{2(k+h^\vee)}\sum_a {:}J^aJ^a{:}$, central charge $c = k\dim\mathfrak{g}/(k+h^\vee)$.

KZ equations. Conformal blocks satisfy

$$\Big[(k+h^\vee)\partial_{z_i} + \sum_{j\neq i}\tfrac{\Omega_{ij}}{z_i - z_j}\Big]F = 0, \quad \Omega_{ij} = \sum_a t^a_i \otimes t^a_j.$$

Monodromy → braid group representations → 3d CS theory.

Verlinde formula.

$$\dim V_{g,n}(\lambda_1,\dots,\lambda_n) = \sum_\mu (S_{\mu 0})^{2-2g}\prod_i \tfrac{S_{\lambda_i\mu}}{S_{0\mu}},$$

where $S_{\lambda\mu}$ is the modular S-matrix. Fusion coefficients $N^\nu_{\lambda\mu} = \sum_\rho S_{\lambda\rho}S_{\mu\rho}\overline{S_{\nu\rho}}/S_{0\rho}$ equal CS conformal block dimensions on a three-holed sphere.

Atiyah's TQFT Axioms

A TQFT of dimension n is a symmetric monoidal functor

$$Z : \mathbf{Cob}_n \longrightarrow \mathbf{Vect}_{\mathbb{F}},$$

assigning: $Z(\Sigma) \in \mathbf{Vect}$ to closed $(n-1)$-manifolds; $Z(M) : Z(\Sigma_0) \to Z(\Sigma_1)$ to cobordisms. Axioms: $Z(\bar\Sigma) = Z(\Sigma)^*$; $Z(\Sigma_0 \sqcup \Sigma_1) = Z(\Sigma_0)\otimes Z(\Sigma_1)$; gluing = composition. Closed M gives partition function $Z(M) \in \mathbb{F}$. For $n = 2$: $Z(S^1)$ is a commutative Frobenius algebra.

Reshetikhin–Turaev TQFT. $Z_{\mathrm{RT}}(\Sigma) = \mathcal{V}(\Sigma, G, k)$ (Verlinde bundle sections); $Z_{\mathrm{RT}}(M) = \int \mathcal{D}A\, e^{iS_{\mathrm{CS}}}$, a rigorous link invariant (Jones polynomial for $G = SU(2)$) built from the modular tensor category of integrable $\hat{\mathfrak{g}}_k$-modules.

Cobordism hypothesis (Baez–Dolan, Lurie). Fully extended framed TQFTs $Z : \mathbf{Cob}^{\mathrm{fr}}_n \to \mathcal{C}$ are classified by $Z(\mathrm{pt}) \in \mathcal{C}$ fully dualizable.

Morse Theory

Morse function $f : M \to \mathbb{R}$, index $\lambda(p) = \#\{\text{neg. eigenvalues of } \mathrm{Hess}_p f\}$. Handle decomposition: crossing $f(p)$ attaches a λ-handle; $M = h^0 \cup \cdots \cup h^n$. Morse complex:

$$C_k = \bigoplus_{\lambda(p)=k}\mathbb{Z}\langle p\rangle, \quad \partial p = \sum_{\lambda(q)=k-1}\#\mathcal{M}(p,q)\cdot q,$$

with $HM_*(M) \cong H_*(M,\mathbb{Z})$. TQFT interpretation: handles ↔ linear maps; 0-, 1-, 2-handles give unit, multiplication, counit of Frobenius algebra.

Floer theory. Chern–Simons functional on $\mathcal{A}(Y)/\mathcal{G}$; critical points = flat connections; gradient flow lines = ASD instantons on $Y\times\mathbb{R}$ ($F^+_A = 0$). Floer homology $HF_*(Y,G)$ is a 3-manifold invariant. Atiyah–Floer conjecture: $HF_*(Y) \cong HF_{\mathrm{Lag}}(\mathcal{L}_1,\mathcal{L}_2)$ for Heegaard splitting $Y = H_1 \cup_\Sigma H_2$.

BV Formalism

BV space $\mathcal{F}_{\mathrm{BV}} = T^*[-1]\mathcal{F}_{\mathrm{ext}}$ carries the *antibracket* (odd degree +1 Poisson bracket)

$$(F,G) = \int\Big(\tfrac{\vec\delta F}{\delta\phi}\tfrac{\delta G}{\delta\phi^*} - \tfrac{\vec\delta F}{\delta\phi^*}\tfrac{\delta G}{\delta\phi}\Big)d^nx.$$

CME: $(S_{\mathrm{BV}}, S_{\mathrm{BV}}) = 0$ encodes gauge invariance, closure, higher Jacobi. BRST operator $s = (S_{\mathrm{BV}},\cdot)$, $s^2 = 0$. **QME:** $\frac{1}{2}(S,S) - i\hbar\Delta S = 0$ with BV Laplacian $\Delta = \int \frac{\vec\delta}{\delta\phi}\frac{\vec\delta}{\delta\phi^*}$, $\Delta^2 = 0$. Gauge fixing: Lagrangian $\mathcal{L}_\Psi = \{\phi^* = \partial\Psi/\partial\phi\}$; $Z = \int_{\mathcal{L}_\Psi}\mathcal{D}\phi\, e^{iS|_{\mathcal{L}_\Psi}/\hbar}$ is Ψ-independent iff QME holds.

CS in BV. Field space $\mathcal{F}^{\mathrm{CS}}_{\mathrm{BV}} = \Omega^\bullet(Y,\mathfrak{g})[1]$ graded by form degree minus 1:

$$S^{\mathrm{CS}}_{\mathrm{BV}} = \tfrac{k}{4\pi}\int_Y \mathrm{tr}\big(\mathbb{A}\wedge d\mathbb{A} + \tfrac{2}{3}\mathbb{A}^3\big).$$

CME ⇔ Maurer–Cartan equation (Jacobi identity). BV-BRST cohomology at $\partial Y = \Sigma$ recovers the WZW chiral algebra. Anomaly criterion: theory is anomaly-free iff $[\Delta S_{\mathrm{BV}}] = 0$ in $H^\bullet(\mathcal{O}^{\mathrm{cl}}, (S,\cdot))$ — vanishes for CS by Kontsevich.